Roque Enrique López Condori

Curriculum Vitae

Personal Information

Name	Roque Enrique López Condori
Nationality	Peruvian
Website	`https://roquelopez.com`
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G. Scholar	`https://scholar.google.com/citations?user=eCBvlNMAAAAJ`

Education

2013–2015 **MSc in Computer Science**, *São Paulo University*, Brazil.

2006–2010 **BSc in Computer Science**, *San Agustin National University*, Peru.

Awards

- 2023, Distinguished participation in the External Operations Platform Data Challenge (JIATF), USA.
- 2023, Best Paper Honorable Mention Award at IEEE Visualization Conference (VIS), Australia.
- 2023, Best Paper Award at Data Management for End-to-End Machine Learning (DEEM) Workshop, co-located with SIGMOD, USA.
- 2020, Finalist team for the Wells Fargo Campus Analytics Challenge 2020, USA.
- 2016, Finalist for the Best MSc Dissertation in IEEE Latin American Conference on Computational Intelligence, Colombia.
- 2014, Young Scientific Best Paper in VII International Conference on Intelligent Information and Engineering Systems, Poland.
- 2014, Master Scholarship granted by Samsung-Brazil (2014-2015), Brazil.
- 2013, Top 5th Bachelor thesis in Simposio Peruano de Inteligencia Artificial, Peru.
- 2010, Placed in top 5th in the undergraduate, Peru.
- 2005, First place and scholarship High School Diploma, Peru.

Experience

Dec 2018 – present **Research Engineer**, *New York University*, USA.

I have worked on multiple DARPA and ARPA-H research projects at NYU's Visualization, Imaging, and Data Analytics lab (VIDA).

- In the BDF project, I'm working on building a Python toolkit for biomedical data integration using Generative Artificial Intelligence (GenAI) and Natural Language Processing (NLP) techniques.
- In the D3M project, using Reinforcement Learning (RL) methods, I have implemented an automated Machine Learning (ML) system that automatically searches for models and derives end-to-end pipelines that read, pre-process the data, and train the model.
- In the PTG project, I explored and implemented different techniques based on Large Language Models (LLMs) for reasoning tasks, e.g. to identify errors in recipe steps and to extract preconditions and effects of actions in recipes.

Feb 2017 – Aug 2018 **Research Engineer**, *Inria*, France.

I worked on extracting symbolic knowledge about objects and their characteristics from unstructured text (relying on natural language processing, machine learning and machine reading techniques), as well as available ontologies and knowledge on the Semantic Web.

Oct 2016 – Sept 2018 **CTO**, *DoctorCV*.

Peruvian startup (`https://doctor-cv.com/`) which evaluates and proposes improvements in your CV using NLP techniques. It is like having a virtual recruiter expert who will tell what to change in the resume to increase the chances of being called for the job interview.

July 2015 – Jan 2017 **Researcher, Software Engineer**, *Elabora Consultoria*, Brazil.

I researched some information retrieval techniques with machine learning methods and word embedding representations to create hierarchies of terms in collections of patent about different topics. In this project, I used Spark, HBase and Python.

Feb 2014 – July 2014 **Teaching Assistant**, *São Paulo University*, Brazil.

I served as TA to professor Dr. Thiago Pardo for the course *Professional Information in Computer Science*. I was responsible for evaluating and grading quizzes and assignments. In collaboration with another TA, we supported 50 students during discussion sections and office hours.

Oct 2013 – Dec 2013 **Software Engineer**, *Dicionário Criativo*, Brazil.

I implemented a crawler for Twitter in Python. This crawler collected public comments about music, books and films. I also developed a module to normalize these comments (spell checker, etc.). With this data and other resources the database was increased.

Jan 2012 – Dec 2012 **Researcher, Software Engineer**, *Lindexa - Wayra Telefónica*, Peru.

I researched and developed some automatic techniques of opinion mining and sentiment analysis for Spanish language. In addition, I implemented a crawler for Facebook and Twitter in Python. This crawler collected all public comments written on these social networks. Lindexa was a winner startup of the first edition of Wayra Peru.

Jan 2011 – Dec 2011 **Technical Researcher, Developer**, *Cátedra CONCYTEC*, Peru.

In general, we researched techniques to generate extractive summaries of many texts. These summaries selected the most important data in terms of coverage and representativeness. I also researched and implemented text classification and clustering algorithms.

Research Projects

Mar 2024 – present
BDF: Biomedical Data Fabric Toolbox, *New York University*, USA.
The BDF project, supported by the Advanced Research Projects Agency for Health (ARPA-H), aims to streamline the integration of biomedical research data from thousands of sources, addressing challenges posed by incompatible data formats.

Dec 2021 – June 2024
PTG: Perceptually-enabled Task Guidance, *New York University*, USA.
The PTG project, funded by the Defense Advanced Research Projects Agency (DARPA), focused on developing methods and technologies for AI assistants that offered just-in-time visual and audio feedback to support task execution.

Dec 2018 – Mar 2024
D3M: Data-Driven Discovery of Models, *New York University*, USA.
The D3M project, a Defense Advanced Research Projects Agency (DARPA) funded effort, focused on developing automated model discovery systems that empower domain experts without data science expertise to build empirical models of complex real-world processes.

Feb 2017 – Aug 2018
ALOOF: Autonomous Learning of the Meaning of Objects, *Inria*, France.
The ALOOF project aimed to advance the ability of robots and autonomous systems to adapt to dynamic real-world environments by enabling them to learn object meanings from web-based resources.

July 2015 – Feb 2017
Patent Faceted Searches Using Topic Modeling, *Elabora Consultoria*, Brazil.
This project aimed to integrate advanced techniques of information retrieval for making faceted searches over big collections of patents about different topics.

Jan 2014 – June 2015
Semantic Text Processing in Brazilian Portuguese, *Samsung – ICMC/USP*, Brazil.
The overall objective of this project was to advance the state of the art in semantic processing of texts written in Brazilian Portuguese. With these resources, we built mining and summarization texts applications, focused on Internet opinions.

Jan 2011 – Dec 2011
Automatic Summarization of Medical Records using Natural Language Processing techniques, *Cátedra CONCYTEC*, Peru.
The goal of this project was to investigate and apply techniques to generate automatic summaries of many documents. We developed different methods to extract, select and organize the most important information.

Monographs

MSc Thesis
Title: Automatic Aspect-Based Opinion Summarization Methods
Description: In this master's project are presented investigations to generate extractive and abstractive summaries of opinions using an aspect-based approach. Besides using known methods in the area, it also was proposed two new methods for Portuguese language which got the best performance in the experiments.

BSc Thesis
Title: Medical Documents Classification based on Keywords using Semantic Information
Description: In this monograph, it is presented a method to classify medical documents which improves the results of Naive Bayes and Rocchio algorithm. This method, in addition to considering statistical information, taking into account the semantic relatedness between the keywords of medical documents.

Languages

Spanish	Fluent	*native language*
English	Advanced Level	*completed studies*
Portuguese	Advanced Level	*completed studies*

Programming Knowledge

ML	Scikit-learn, Numpy, Pandas
RL	Gymnasium, Ray
NLP	NLTK, PyTorch, SpaCy
GenAI	Transformers, LangChain, MCP
Languages	Python, Java, C++
Others	Git, Docker, Linux

Main Interests

- Data Integration
- Data-centric AI
- Applied Machine Learning
- Natural Language Processing
- Reinforcement Learning
- Generative Artificial Intelligence